

Drone Payload Structure Design and Analysis

In this project, you are optimizing drone arm design, including geometry and material selection, to maximize payload capacity while ensuring structural integrity under motor-induced loads. You will use MATLAB to perform force balance analysis and finite element modeling to quantitatively support your design solution.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this [video](#).

Suggested Tasks

1. Gather your starter assumptions and load the material properties into MATLAB.
2. Propose at least two drone arm designs
3. Perform a thrust-to-weight analysis to evaluate each of your designs across all material options. Your design must meet a minimum payload requirement and safety standards.
4. Perform finite element analysis across all design and material options.
5. Recommend a final design solution.

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Break Down the Problem

In the space below, define the problem statement:

The objective of this project is to test different drone arm designs and material options for a quadcopter drone. The goal is to maximize payload capacity while maintaining structural integrity under motor induced loads. MATLAB will be used to perform thrust-to-weight analysis and finite element analysis to compare each design and material combination, allowing us to choose the best design that meets the design requirements.

List the requirements, constraints, and success criteria for the project:

Requirements

- Design and model two or more drone arms for a quadcopter drone
- With given material options, perform thrust to weight analysis across each option to see which material optimizes payload capacity while meeting safety requirements
- Perform finite element analysis on each drone arm's 3D model in the form of an .stl file in MATLAB from a combination of upward motor thrust force and downward motor weight force to obtain its maximum von Mises stress, displacement, and safety factor.
- Compare and contrast material options to see which arm meets the design requirements based on the Von mises stresses, displacements, and safety factor from the FEA

Constraints

- Safety factor must be at least 1.5
- Assuming 1kg of thrust per motor, drone must be able to support a minimum 0.5kg payload
- Thrust to weight ratio must be at least 2:1

Success Criteria

- At least two drone design sketches showing key geometric features
- Results of thrust-to-weight analysis presented either in a summary table or plot
- Results of finite element analysis
- Numerical values presented in a summary table
- Visualizations of x-, y-, and z-displacement and Von Mises stress
- A final design recommendation justified using the results of your analyses

What is your proposed solution?

We propose to make a MATLAB live script that can automatically analyze multiple drone arm designs in the form of .stl files, allowing the user to know each design's maximum payload, average material cost, and structural integrity under motor-induced loads for each user-defined material combination.

List the steps you will need to take to achieve your proposed solution:

1. Gather starter assumptions and define material properties into MATLAB
2. Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features
3. Perform a thrust-to-weight analysis across all design and material options to evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements
4. Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. (To simplify the task, we will perform the FEA only on a single drone arm.)
5. Using the results of both the thrust-to-weight analysis and finite element analysis, propose a final design solution for the drone arm that maximizes payload capacity while meeting safe flight standards and maintaining structural integrity under load

What challenges do you face in the process of achieving the solution?

- Creating a real-life scale drone arm in CAD
- Creating a user-interactable live script in MATLAB that meets all of the design requirements
- Presenting data in a user-friendly format in MATLAB
- Troubleshooting code issues
- Coordinating with teammates and effectively using GitHub

Task 1: Gather your starter assumptions and load material properties

Load the provided `droneArmMaterials.mat` file into MATLAB. This contains a data struct with the material properties for six different materials commonly used for drones (carbon fiber composite, aluminum alloy, fiberglass composite, PLA plastic, ABS plastic, and wood). For each material, you have been provided the following information:

- Density (ρ), in $\frac{\text{kg}}{\text{m}^3}$
- Young's Modulus (E) in Pa
- Poisson's ratio (ν), unitless
- Yield strength, in Pa
- Average cost per meter in US dollars

For this project, you can assume the following design parameters:

- Quadcopter drone with 4 motors and 4 propellers
- Each motor can provide up to 1kg thrust
- Acceleration due to gravity: $g = 9.81 \frac{\text{m}}{\text{s}^2}$
- Total weight of the drone components without the arms or the payload: 1kg

The breakdown of the weight of each drone component is as follow:

- Battery = 0.3kg
- Each motor = 65g, 4 motors = 0.26kg
- Each propeller = 10g, 4 propellers = 0.04 kg
- Each electronic speed controller = 30g, 4 electronic speed controllers = 0.12 kg
- Flight controller and electronics = 0.05kg
- Receiver/GPS = 0.03kg
- Drone frame (center body) excluding arms = 0.2kg

Design Parameters

Before you run the code, make sure your current directory is set to the project folder!

```
%Press "Hide Code" from the top menu bar's view setting for the best experience.
```

```
%Initialize stuff for running this code cleanly
```

```

clear; %Clears all variables from memory
clc; %Clears command window

tic

addpath(genpath('resource')); %Adds relative file path of project resources
load('droneArmMaterials.mat'); %Loads material property data; Make sure
your working directory is set to the project root (use command cd in
command window)!

materials = struct2table(materials); %Converts the structure array to a
table to allow for easier vectorization

%Initialize global variables
g = 9.8; % acceleration due to gravity, m/s^2

%Initialize drone design parameters
requiredThrustToWeightRatio = 2; %Drone's required thrust to weight ratio
baseDroneMass_kg = 1; %Target drone body's mass excluding arms and payload
in kg
minimumPayload_kg = 0.5; %Drone's minimum payload capacity in kg

numArms = 4; %Total number of arms on drone
numMotors = numArms; %Total number of motors on drone (one motor per arm)
numPropellers = numArms; %Total number of propellers on drone (one
propeller per motor)

maxThrustPerMotor_kg = 1; %Mass of thrust per motor in kg
totalThrust_kg = numMotors * maxThrustPerMotor_kg; %Maximum thrust mass
from all of a drone's motors in kg

maxThrustPerMotor_N = maxThrustPerMotor_kg * g; %Thrust force per motor in N
totalMaxThrust_N = numMotors * maxThrustPerMotor_N; %Maximum thrust force
from all of a drone's motors in N

```

Drone Component Masses

```

%Initialize drone specs
batteryMass_kg = 0.3; %Drone's battery mass in kg
motorMass_each_kg = 0.065; %Each of the drone's motors' mass in kg
propellerMass_each_kg = 0.01; %Each of the drone's propellers' mass in kg
escMass_each_kg = 0.03; %Each of the drone's electronic speed controllers'
mass in kg
flightControllerMass_kg = 0.05; %Drone's flight controller and its
electronics' mass in kg
receiverGPSSMass_kg = 0.03; %Drone's GPS reciever unit's mass in kg
centerFrameMass_kg = 0.2; %Drone frame's mass in kg

baseMassCheck_kg = batteryMass_kg + ...
numMotors*motorMass_each_kg + ...

```

```

numPropellers*propellerMass_each_kg + ...
numMotors*escMass_each_kg + ...
flightControllerMass_kg + ...
receiverGPSSMass_kg + ...
centerFrameMass_kg; %Add together all the drone's
components' mass in kg

```

Key Values

```

%Display key values
fprintf("Total maximum thrust = %.2f N\n", totalMaxThrust_N);

```

Total maximum thrust = 39.20 N

```

fprintf("Base drone mass check = %.2f kg\n", baseMassCheck_kg);

```

Base drone mass check = 1.00 kg

```

fprintf("Minimum payload requirement = %.2f kg\n", minimumPayload_kg);

```

Minimum payload requirement = 0.50 kg

```

fprintf("Required thrust-to-weight ratio = %.1f:1\n",
requiredThrustToWeightRatio);

```

Required thrust-to-weight ratio = 2.0:1

Task 2: Propose at least two drone arm designs

Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features, including:

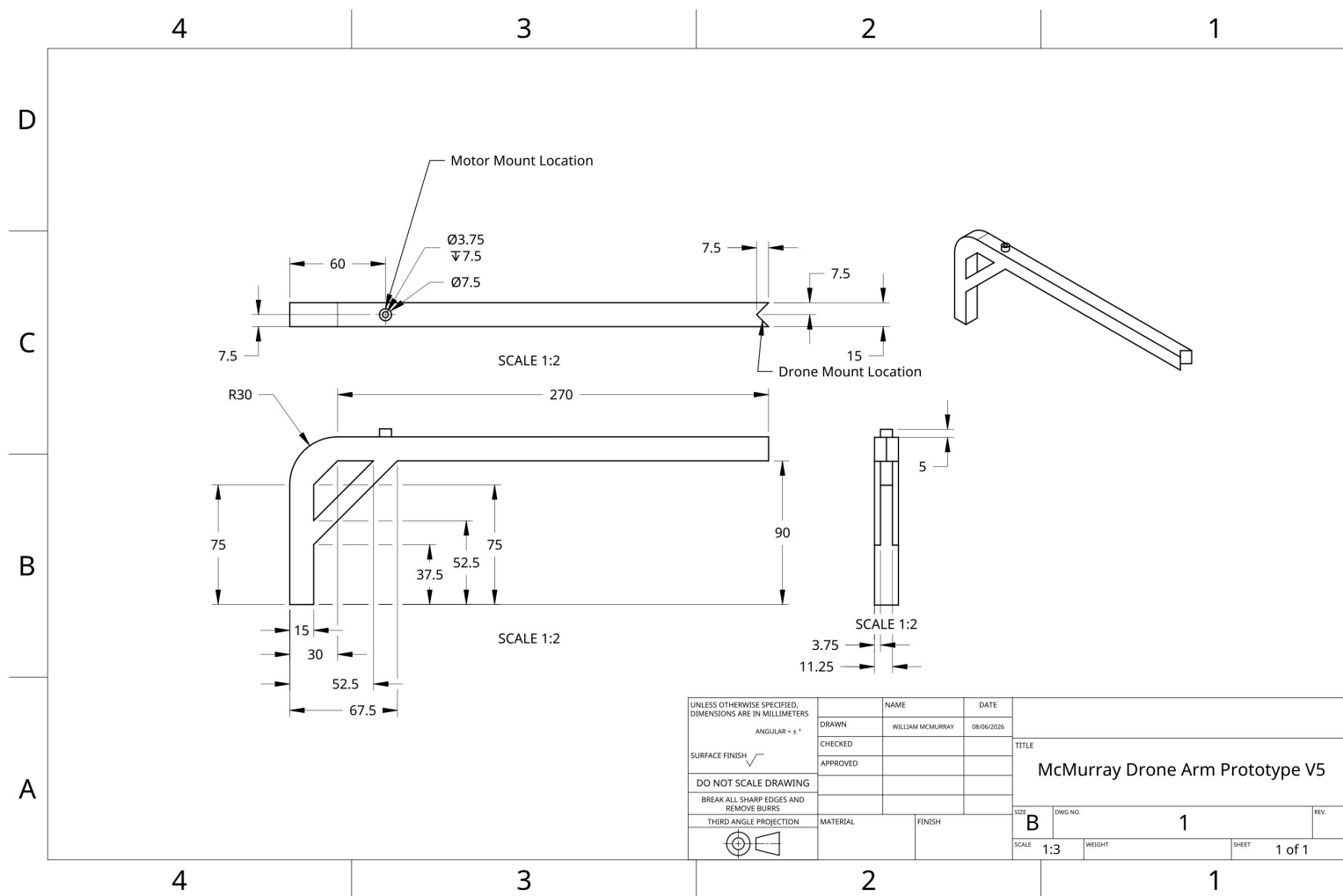
- Overall shape (e.g. beam, tube, truss structure)
- Dimensions (length, width, thickness)
- Motor mount location
- Where the arm connects to the drone body

The two designs should be meaningfully different from each other (e.g. different geometries or structural strategies) and not just small variations of the same idea.

Navigate to the 'Insert' tab and insert pictures of your sketches in the space below (hand-drawn or digital). Include a short design description for each sketch that includes:

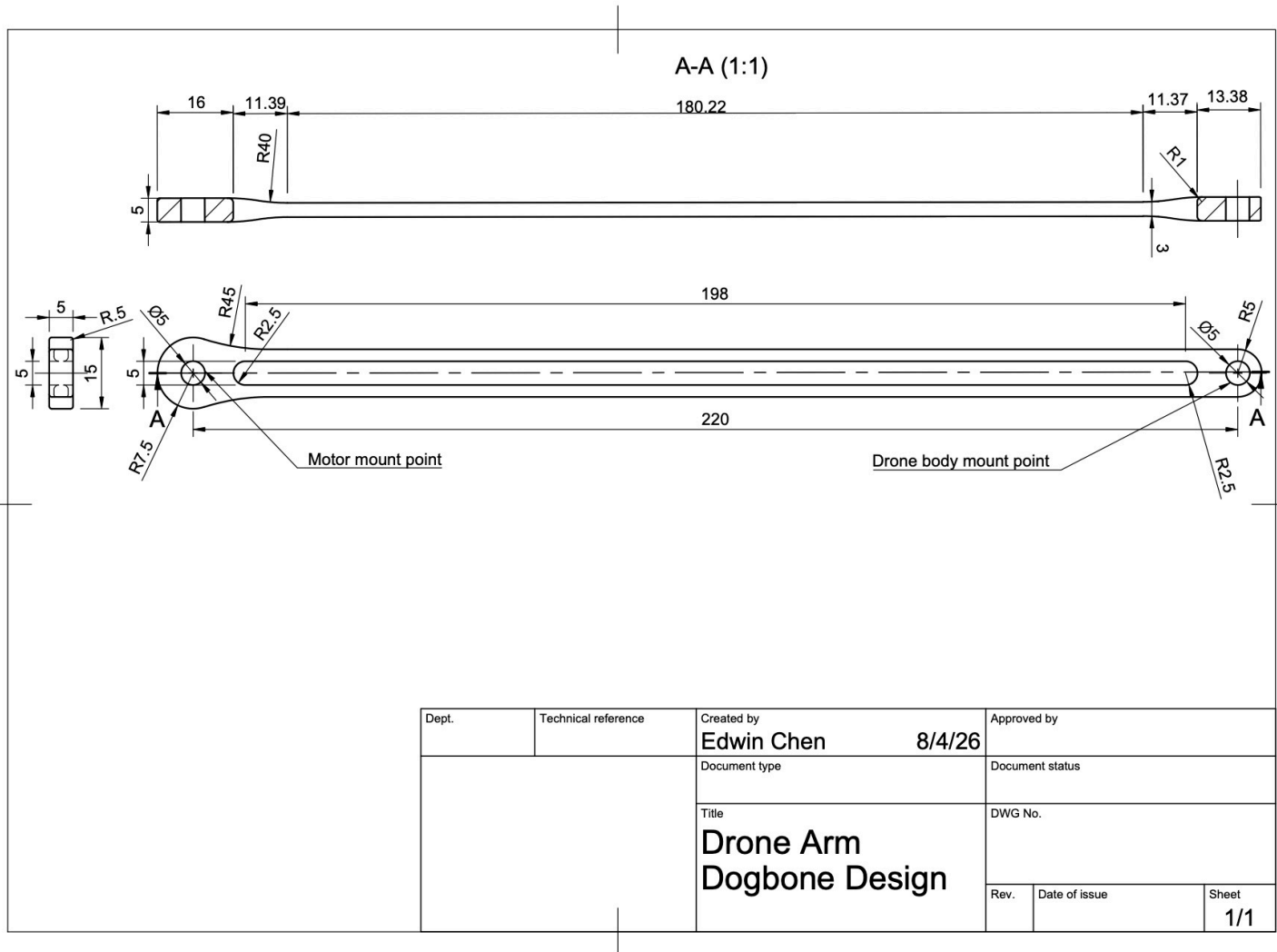
- Geometry (e.g. solid beam, hollow tube, beam with trusses)
- Why you chose this design
- How it might perform under load

McMurray_DroneArmPrototypeV5



My design is a long drone arm that bends down into the leg with a fillet on the outside and a chamfer on the inside in order to remove and distribute any additional stress that might result from sharp corners. An additional pair of ribs connect the arm and leg for additional stability to support the arm that will attach to the drone holding the payload. The motor mount extends out from the arm cylindrically and has a smaller 3.75mm cylindrical hole for the motor to be placed in. The drone arms will mount to each of the four corners of a square shaped drone body and will extrude diagonal to each face of the body. Under the load of the motor, the diagonal beam connecting the arm and the leg will take away some of the stress from going to the long arm. However, PLA plastic will provide the least effective arm in terms of displacement because of its high elasticity and low strength. -Will

chris_final_drone_arm



This drone arm design is a narrow and solid beam with dimensions similar to what's expected for a fast-flying FPV drone. I chose this design because the cut and slimming taper in the middle will save weight and material, while the motor and drone body mount points' has extra material to increase the amount of threading available and act as additional reinforcement to improve long-term durability (The dogbone name comes from the middle taper in the drone arm's side profile). Sharp corners were eliminated through fillet operations to decrease stress concentrations, giving it a round design. Even so, this arm design will suffer from bending and torsional stresses under a motor's load because of the lack of material and bracing in the arm's tapered portion. Therefore, this design warrants the usage of a very stiff material like a carbon-fiber reinforced polymer composite to minimize the bending and torsional stresses and maximize its effectiveness. -Edwin

Task 3: Perform thrust-to-weight analysis across all design and material options

Write a MATLAB function that will perform a thrust-to-weight analysis across all design and material options. This analysis will help you evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements.

- Your design must support a minimum payload of 0.5kg.
- Your design must meet a thrust-to-weight ratio of at least 2:1 to ensure safe and stable flight. This safety standard is widely used in drone design. Conceptually, it means that the drone's maximum thrust must be at least twice its weight. This ensures stable hovering and maneuvering with a safety margin. Note that the thrust-to-weight safety margin is a system-level safety standard for drone flight that is distinct from the structural safety factor of the drone arms (which you will evaluate later through finite element analysis).
- Your goal is calculate maximum payload capacity for each design and material option (i.e. what is the heaviest payload your drone design could carry while still allowing the drone to safely take off?)

Ensure that the results of your analysis can be readily interpreted, e.g. in the form of a summary table or plot.

Note: if you have two drone arm designs, the expected output should be 12 payload capacity estimates (2 designs across 6 materials options).

```
% Insert your code in the spaces below (or make use of helper functions and
% indicate where
% they can be found). Ensure your code is well-documented.
```

```
% Hint: General suggested steps for the thrust-to-weight analysis-
% 1. Start with the known values (motor thrust capability and base drone
% mass without the payload). Note that for the base drone mass, you will
% need to calculate weight estimations for each proposed drone arm design
% and each material option.
% 2. Add payload as a variable component of the weight.
% 3. Apply the requirement that total thrust must be at least twice the
% total weight of the drone.
% 4. Solve for maximum payload capacity
```

```
% Maximum total drone mass allowed by 2:1 thrust-to-weight ratio
maxAllowedTotalMass_kg = totalThrust_kg / requiredThrustToWeightRatio;
```

```
fprintf("Maximum allowed total drone mass = %.2f kg\n",
maxAllowedTotalMass_kg);
```

```
Maximum allowed total drone mass = 2.00 kg
```

Make sure all the files you want to analyze is located in the resource/stl folder!

```
filePath = 'resource/stl'; %Specify the file path for the program to search
for the .stl files. I don't recommend changing this unless absolutely
necessary.
```

```
%Calculate the maximum payload of each material for each drone arm design
```

```
fileInfo = dir(fullfile(filePath, '*.stl')); %Search for all of the file
names in the resource stl folder
```

```

fileName = string({fileInfo.name}); %Create an array of all of the
available file names

HParameters = [5, 0.001]; %[Hmax, Hmin]; Hmax is the maximum edge length
and Hmin is the minimum edge length to be used to generate a PDE mesh.
I don't recommend changing these values because MATLAB might refuse to
generate a mesh.

for i = 1:length(fileName)
    fprintf("Analyzing " + fileName(i) + "...");
    model_fe = femodel(AnalysisType="structuralStatic",
Geometry=fileName(i)); %Initializes the finite element model of the
selected stl file
    model_mesh = createpde; %Initializes a partial differential equation
geometry
    importGeometry(model_mesh, fileName(i)); %Initializes the partial
differential equation geometry from the selected stl file
    model_mesh = generateMesh(model_mesh, Hmax = HParameters(1), Hmin =
HParameters(2)); %Initializes a mesh made of tris from the selected stl
file.

    %Create a table of payload data for every drone arm model
    armMass(:,i) = materials.rho_kg_m3 * volume(model_mesh); %Mass of one
drone arm; STL units are in m, so volume is m^3
    payloadCapacity(:,i) = maxAllowedTotalMass_kg - baseDroneMass_kg -
(numArms .* armMass(:,i)); % Maximum payload capacity of the drone based on
the material
    Design_Cost_USD(:,i) = materials.cost_USD_per_m .*
(volume(model_mesh))^(1/3); %How much the design would cost per material

```

Thrust-to-Weight Analysis Results

```

resultsTable(i) = {table(materials.name, armMass(:,i), numArms .*
armMass(:,i), payloadCapacity(:,i), Design_Cost_USD(:,i),...
    'VariableNames',
{'Material','OneArmMass_kg','TotalArmMass_kg','payloadCapacity_kg','Design_C
ost_USD'})}; %Create a table summary of data for an stl model
resultsTable{i} %Display this iteration's table

%Plot payload capacity of each arm design and material combination
figure %Initialize figure

bar(resultsTable{i}.payloadCapacity_kg) %Create bar graphs for the max
payload values for each different material
xticklabels(materials.name) %Define name of bar graphs
xtickangle(45) %Rotate the bar graph names by 45 degrees

ylabel("Maximum Payload Capacity (kg)") %y-axis label
xlabel("Material") %x-axis label

```

```

    chartTitle = fileName(i) + ": Payload Capacity by Material"; %Create
custom title for each figure through string concatenation
    title(chartTitle); %Add custom title to current figure

    grid on

    %Plot cost for the drone arm per material
    figure %Initialize figure

    bar(resultsTable{i}.Design_Cost_USD) %Create bar graphs for the cost
for each different material
    xticklabels(materials.name) %Define name of bar graphs
    xtickangle(45) %Rotate the bar graph names by 45 degrees

    ylabel("Design Cost ($USD)") %y-axis label
    xlabel("Material") %x-axis label

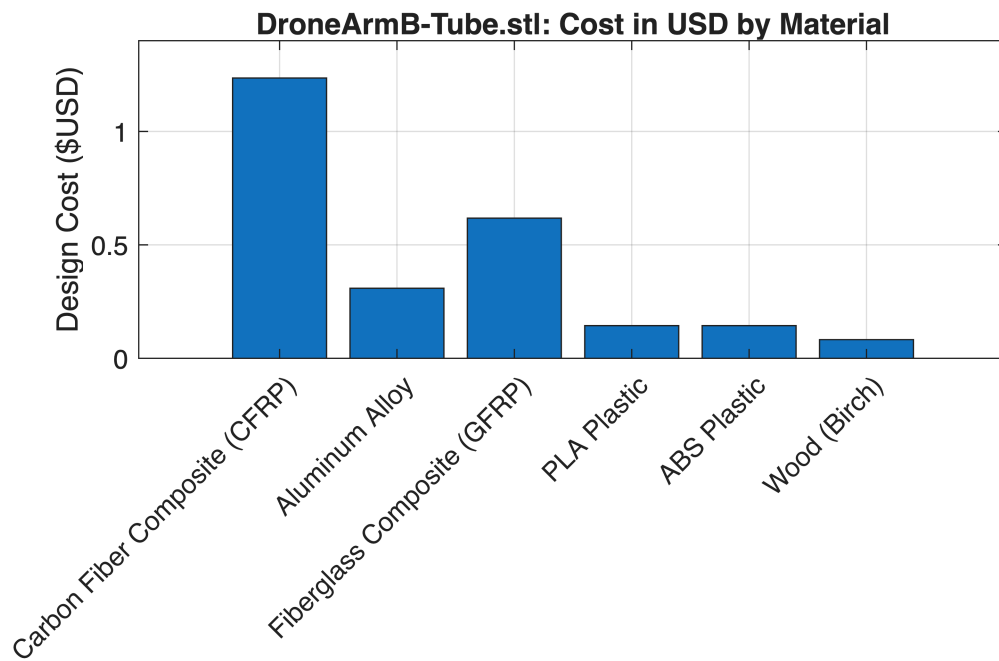
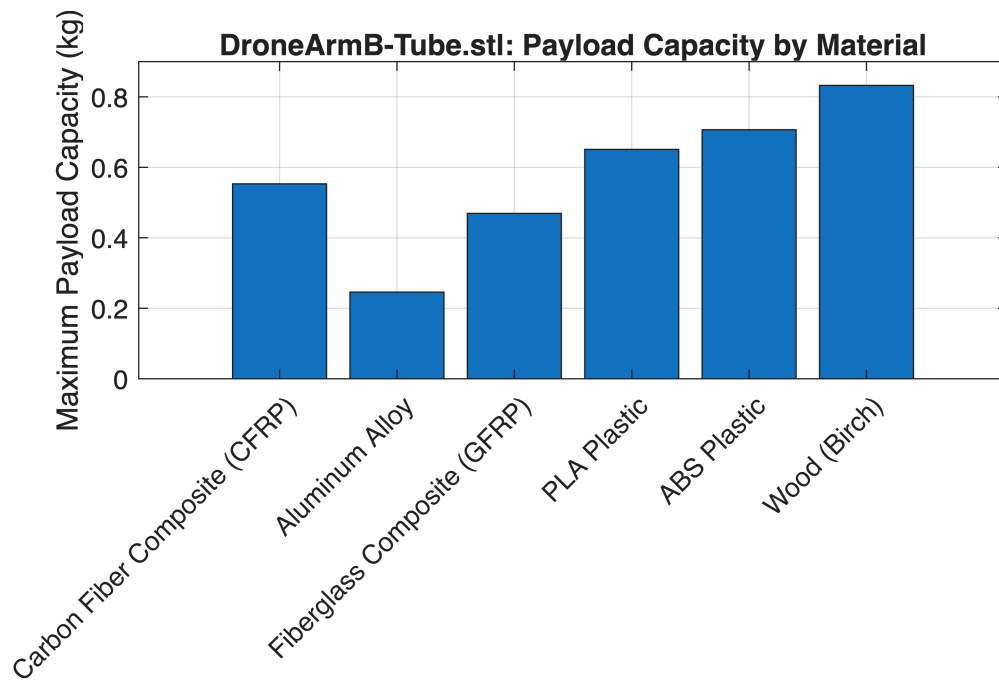
    chartTitle = fileName(i) + ": Cost in USD by Material"; %Create custom
title for each figure through string concatenation
    title(chartTitle); %Add custom title to current figure

    grid on
end

```

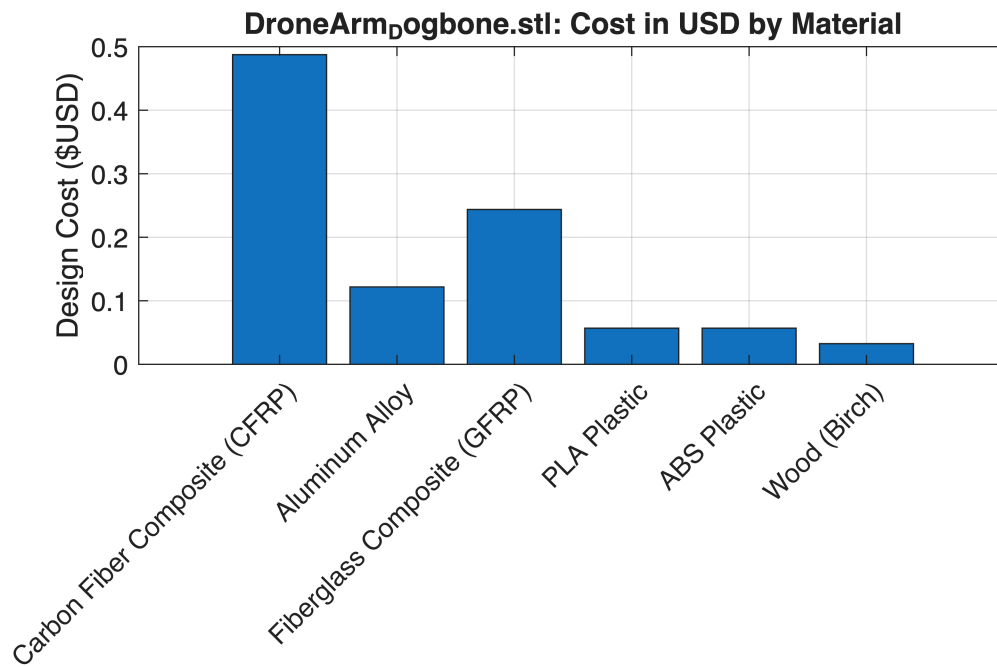
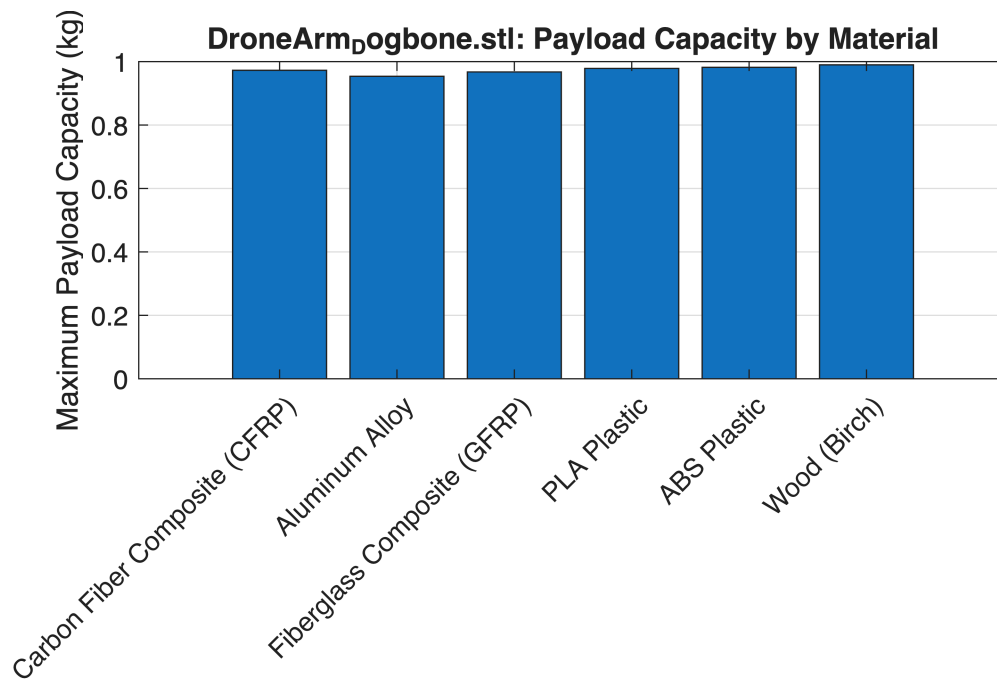
Analyzing DroneArmB-Tube.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.1117	0.4470	0.5530
2	"Aluminum Alloy"	0.1886	0.7543	0.2457
3	"Fiberglass Composite (GFRP)"	0.1327	0.5308	0.4692
4	"PLA Plastic"	0.0873	0.3492	0.6508
5	"ABS Plastic"	0.0733	0.2933	0.7067
6	"Wood (Birch)"	0.0419	0.1676	0.8324



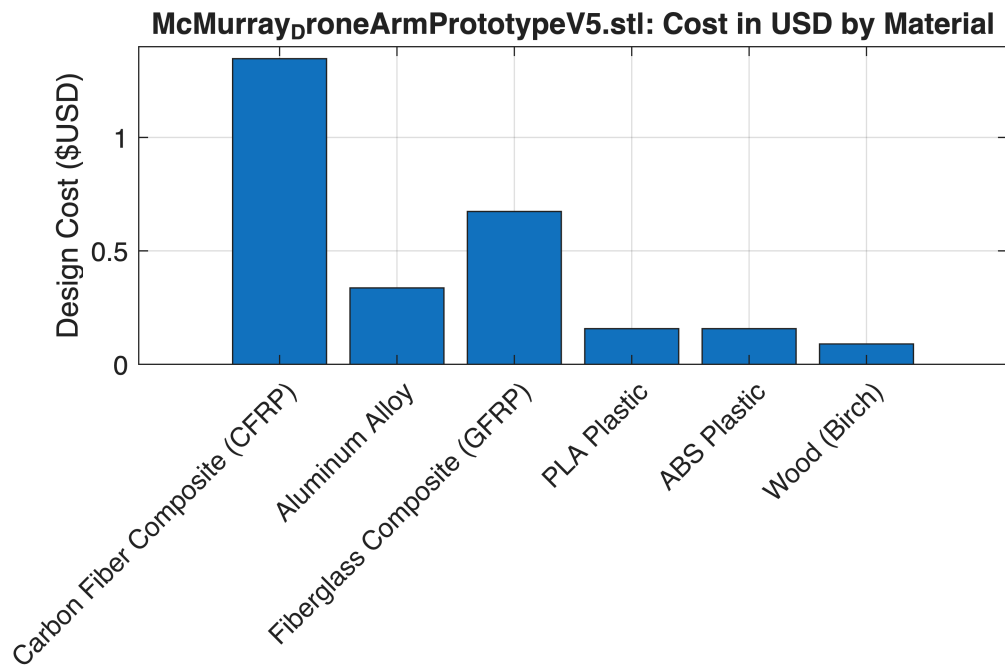
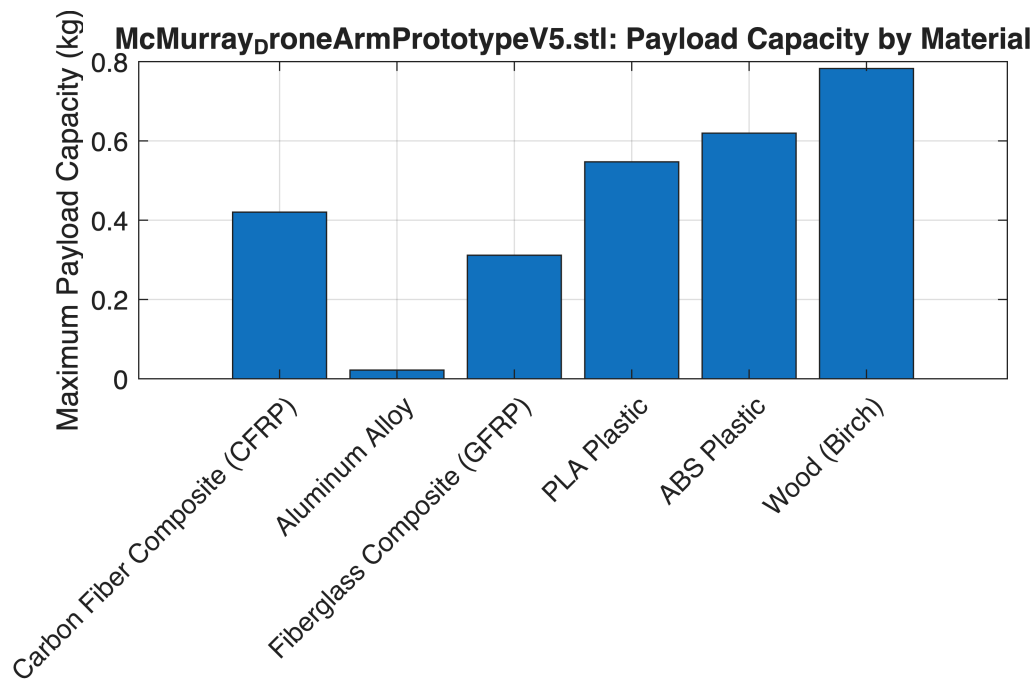
Analyzing DroneArm_Dogbone.stl...
ans = 6x5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.0069	0.0274	0.9726
2	"Aluminum Alloy"	0.0116	0.0463	0.9537
3	"Fiberglass Composite (GFRP)"	0.0081	0.0326	0.9674
4	"PLA Plastic"	0.0054	0.0214	0.9786
5	"ABS Plastic"	0.0045	0.0180	0.9820
6	"Wood (Birch)"	0.0026	0.0103	0.9897



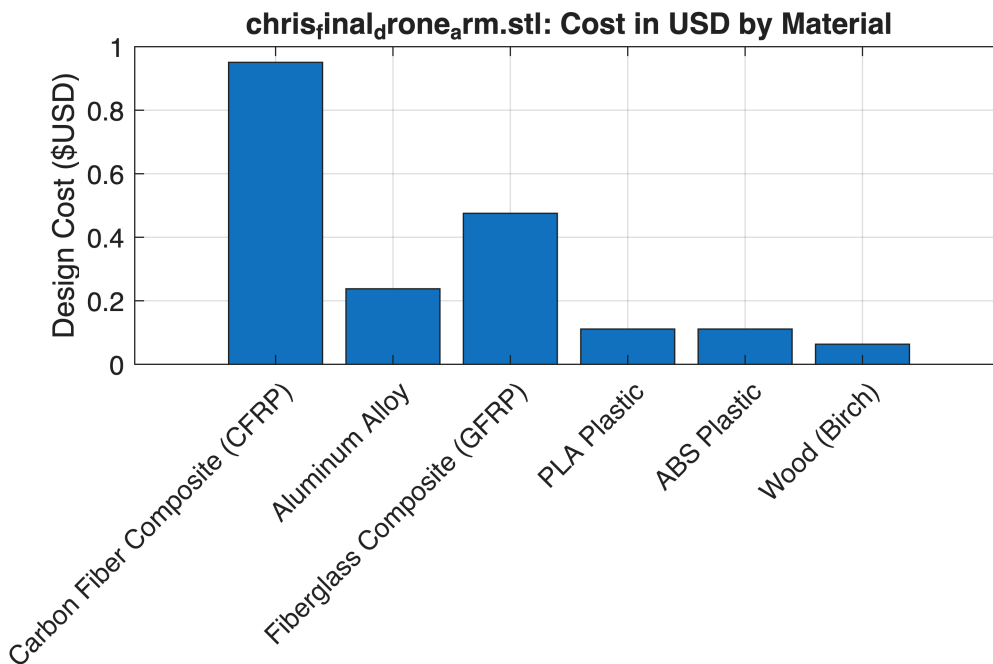
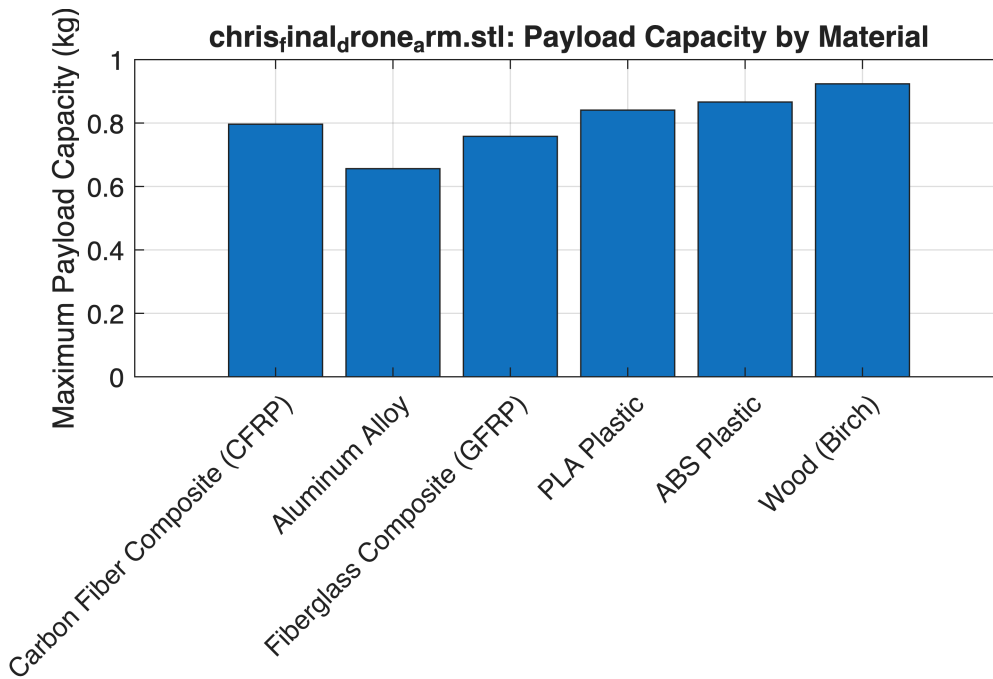
Analyzing McMurray_DroneArmPrototypeV5.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.1449	0.5796	0.4204
2	"Aluminum Alloy"	0.2445	0.9781	0.0219
3	"Fiberglass Composite (GFRP)"	0.1721	0.6883	0.3117
4	"PLA Plastic"	0.1132	0.4528	0.5472
5	"ABS Plastic"	0.0951	0.3804	0.6196
6	"Wood (Birch)"	0.0543	0.2174	0.7826



Analyzing chris_final_drone_arm.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.0509	0.2037	0.7963
2	"Aluminum Alloy"	0.0860	0.3438	0.6562
3	"Fiberglass Composite (GFRP)"	0.0605	0.2419	0.7581
4	"PLA Plastic"	0.0398	0.1592	0.8408
5	"ABS Plastic"	0.0334	0.1337	0.8663
6	"Wood (Birch)"	0.0191	0.0764	0.9236



Task 4: Perform finite element analysis across all design and material options

Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. To simplify the task, you will perform the FEA only on a single drone arm.

Using a CAD software of your choice, first create a 3D model of one drone arm for each of your designs. Be sure your model can be exported as a STEP or STL file for analysis in MATLAB.

- Recommended CAD software options: Tinkercad, Onshape, Fusion 360, Solidworks (most have free versions)

Import your model into MATLAB and perform FEA. In your FEA model, you should model and apply two loads to the drone arm: upward thrust force from the motor and downward weight of the motor.

- Helpful resources: [online tutorial](#), documentation for [Structural Mechanics](#) and [Partial Differentiation Equation Toolbox](#), and [example workflow](#)

For each design and material combination, you should report the overall maximum displacement, maximum stress, and factor of safety and include visualizations of the x-, y-, and z-displacement and Von Mises stress. Ensure that your results are presented for easy interpretation (e.g. present a table with numerical results, clearly labeled subplots for the visualizations).

- The factor of safety compares how strong a material is to how much stress it experiences, indicating whether a design is safe under the given loads. For this project use the equation:

$$FoS = \frac{\text{Material Yield Strength (Pa)}}{\text{Maximum von Mises Stress}}$$

Yield strength indicates how much stress a given material can withstand before permanently deforming. Maximum von Mises stress (derived from your FEA model) is a measure of how much stress your design experiences under the applied loads.

- A typical engineering safety margin for FoS is ~1.5-2+
- This [example](#) demonstrates the plots you are asked to include in your results

Choose the file to perform FEA on

```
fileNameFEA = fileName(4); %Choose a file to perform FEA

%Reinitialize the models for FEA
model_fe = femodel(AnalysisType="structuralStatic", Geometry=fileNameFEA);
%Initializes the finite element model of the selected stl file
model_mesh = createpde; %Initializes a partial differential equation
geometry
importGeometry(model_mesh, fileNameFEA); %Initializes the partial
differential equation geometry from the selected stl file
model_mesh = generateMesh(model_mesh, Hmax = HParameters(1), Hmin =
HParameters(2)); %Initializes a mesh made of tris from the selected stl
file.
```

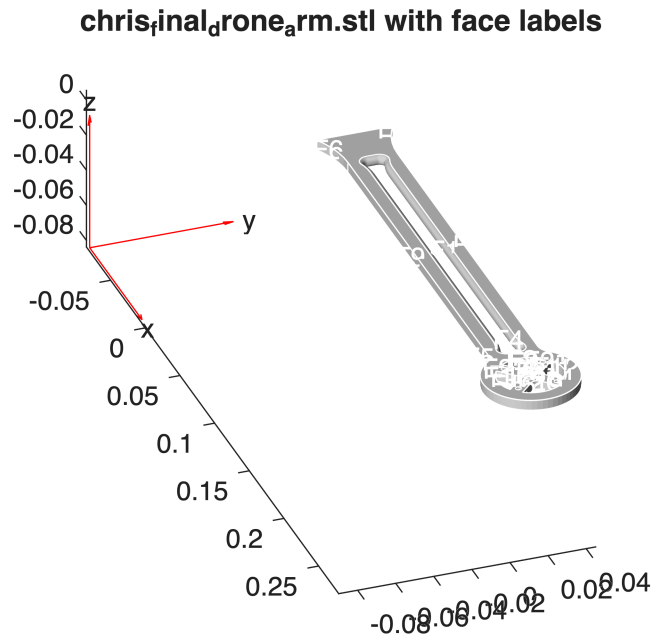
Modify view angle of visualizer

```
% Import drone arm geometry into MATLAB, visualize the geometry to identify
key faces (e.g. tip face)
viewAzimuth = 70;
viewElevation = 30;
viewAngle = [viewAzimuth, viewElevation];
figureTitle = fileNameFEA + " with face labels";

figure
```



```
pdegplot(model_fe,FaceLabels='on') %Plots the finite element model with the
specified view angle with face labels
view(viewAngle);
title(figureTitle)
```



View the visualized model and identify the static and loaded face

```
% Apply boundary conditions (fix base of arm where it would be attached to
drone body)
staticFace = 7; %Specifies the face or boundary condition where the drone
arm will be attached to the drone body
model_fe.FaceBC(staticFace) = faceBC(Constraint="fixed"); %Initializes the
static face where the drone arm will be attached to the drone body in the
finite element model
```

```
% Apply the upward thrust force (motor force); Define the thrust
force, convert force to surface traction, apply surface traction to the
appropriate face in the appropriate direction
loadFace = 8; %Specifies the face or boundary condition where the motor
will be attached to the drone body
loadFaceSurfaceArea = computeFaceSurfaceArea(model_mesh,loadFace); %Compute
the effective surface area of the load face
```

Warning: Some input points are not referenced by the triangulation.

```
vectorMotorThrust = [0; 0; maxThrustPerMotor_N]; %Initializes the force
vector of the applied load
```

```
% Apply downward motor weight; Define motor weight, convert to traction,
apply to the appropriate face in the appropriate direction
vectorMotorWeight = [0; 0; -motorMass_each_kg .* g]; %Initializes the force
vector of the motor's weight
```

```

vectorMotor = (vectorMotorThrust + vectorMotorWeight)./loadFaceSurfaceArea;
%Combine the two force vectors and divide by effective surface area to get
a single net traction value

model_fe.FaceLoad(loadFace) = faceLoad(SurfaceTraction=vectorMotor);
%Initializes the loading face where the motor will be attached to the drone
body in the finite element model

% Optional: include the force of gravity on the arm

%model_fe.FaceLoad = faceLoad(Gravity=g); %This implementation breaks the
FEA for some reason.

% Define material properties
for i = 1:length(materials.name) %Do FEM for all 6 materials for one drone
design
    model_fe.MaterialProperties
= materialProperties(YoungsModulus=materials.E_Pa(i),
PoissonsRatio=materials.nu(i), MassDensity=materials.rho_kg_m3(i));
    model_mesh = generateMesh(model_fe, Hmax = HParameters(1), Hmin =
HParameters(2)); %Regenerates the mesh so it will have the FEM's material
properties
    resultsFEA(i) = solve(model_mesh); %Solve the FEA model (using the
solve function) and store the results for analysis
end

% Get numerical results; The results object generated above contain fields
for displacement and stress.
% Report maximum displacement magnitude and maximum Von Mises stress for
each material-design combination.
% Use the maximum Von Mises stress and given material yield strength to
compute factor of safety for each material-design combination.
% Present your results for easy comparison (e.g. summary table or plot).

for i = 1:length(materials.name)
    maxDisplacement(i) = max(resultsFEA(i).Displacement.Magnitude); %Find
the max displacement of the specified material for the specified drone arm
design in m
    maxVMS(i) = max(resultsFEA(i).VonMisesStress); %Find the max von Mises
stress of the specified material for the specified drone arm design in Pa
end

maxDisplacement = maxDisplacement(:); %Converts row vector into column
vector so it can be added to a table later
maxVMS = maxVMS(:); %Converts row vector into column vector so it can be
added to a table later
safetyFactor = materials.yieldStrength_Pa./maxVMS; %Calculates each
material's safety factor for the specified drone arm design using the
formula defined in the task prompt.

```

FEA Results

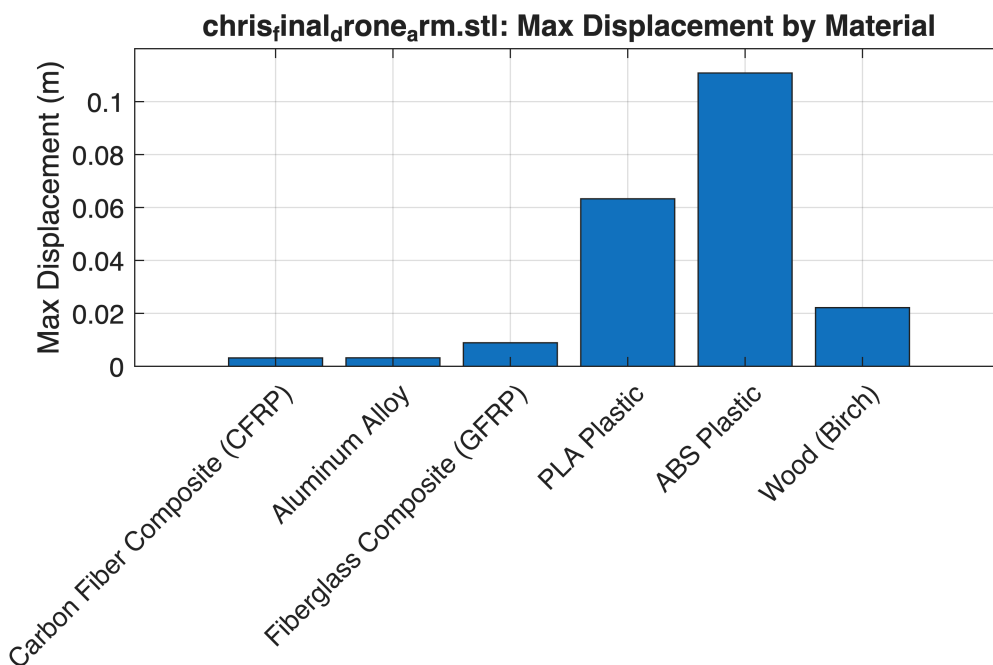
```
%Plot max displacement for the drone arm per material
figure %Initialize figure

bar(maxDisplacement) %Create bar graphs for the cost for each different
material
xticklabels(materials.name) %Define name of bar graphs
xtickangle(45) %Rotate the bar graph names by 45 degrees

ylabel("Max Displacement (m)") %y-axis label
xlabel("Material") %x-axis label

chartTitle = fileNameFEA + ": Max Displacement by Material"; %Create custom
title for each figure through string concatenation
title(chartTitle); %Add custom title to current figure

grid on
```



```
%Plot max von Mises stress for the drone arm per material
figure %Initialize figure

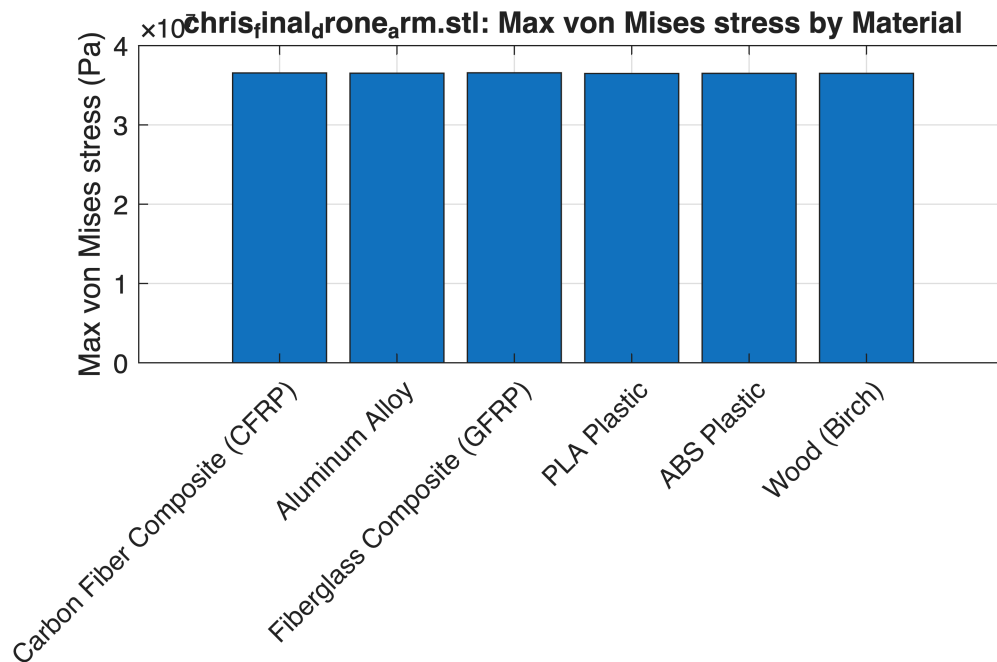
bar(maxVMS) %Create bar graphs for the cost for each different material
xticklabels(materials.name) %Define name of bar graphs
xtickangle(45) %Rotate the bar graph names by 45 degrees

ylabel("Max von Mises stress (Pa)") %y-axis label
```

```
%xlabel("Material") %x-axis label
```

```
chartTitle = fileNameFEA + ": Max von Mises stress by Material"; %Create custom title for each figure through string concatenation
title(chartTitle); %Add custom title to current figure
```

```
grid on
```



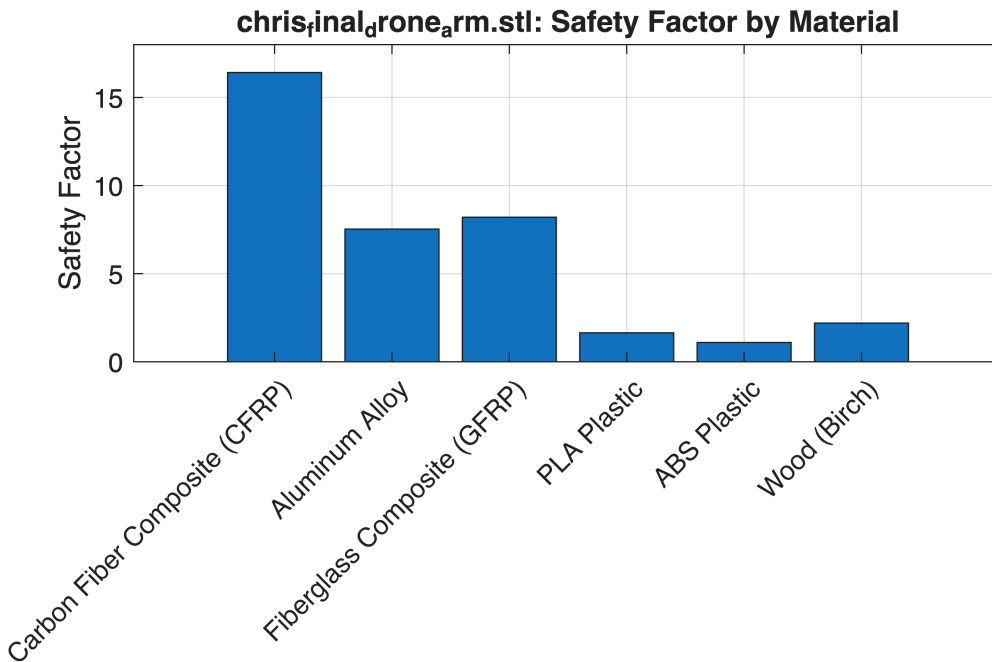
```
%Plot safety factor for the drone arm per material
figure %Initialize figure
```

```
bar(safetyFactor) %Create bar graphs for the cost for each different material
xticklabels(materials.name) %Define name of bar graphs
xtickangle(45) %Rotate the bar graph names by 45 degrees
```

```
ylabel("Safety Factor") %y-axis label
%xlabel("Material") %x-axis label
```

```
chartTitle = fileNameFEA + ": Safety Factor by Material"; %Create custom title for each figure through string concatenation
title(chartTitle); %Add custom title to current figure
```

```
grid on
```

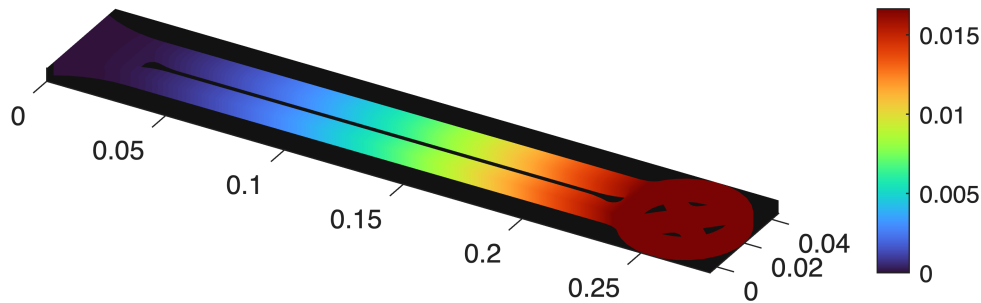


% Visualize your results; Create plots to visualize the x-, y-, and z-displacement and the von Mises stress.
 % Hint: Use the Live Editor task "Visualize PDE Results" or the function pdeplot3D

Select material to visualize the PDE results

```
materialVisual = 6; %Since the Visualize PDE Results task doesn't account
for an array of FEA results (it only shows the first entry), we'll have to
specify which material's results we want to visualize.
resultsFEA_viz = resultsFEA(materialVisual); %Initialize the singular FEA
result for visualization
% Data to visualize
meshData = resultsFEA_viz.Mesh;
nodalData = resultsFEA_viz.Displacement.Magnitude;
deformationData = resultsFEA_viz.Displacement;

% Create PDE result visualization
resultViz = pdeviz(meshData,abs(real(nodalData)), ...
    "DeformationData",deformationData, ...
    "DeformationScaleFactor",0, ...
    "ColorLimits",[0 0.01664]);
```



```
% Clear temporary variables
clearvars meshData nodalData deformationData
```

Final Summary Table

```
idx = find(fileName == fileNameFEA); %Searches for the index of the file
chosen for FEA in the fileName variable to retrieve the correct table for
resultsTable in the next line
resultsFEA_Table = table(materials.name, maxDisplacement,
maxVMS, safetyFactor, resultsTable{idx}.payloadCapacity_kg,
resultsTable{idx}.Design_Cost_USD,...
    'VariableNames',
{'Material', 'maxDisplacement_m', 'maxVonMisesStress_Pa', 'safetyFactor', 'paylo
adCapacity_kg', 'Design_Cost_USD'}) %Create a table summary of data for the
FEA
```

```
resultsFEA_Table = 6x6 table
```

...

	Material	maxDisplacement_m	maxVonMisesStress_Pa
1	"Carbon Fiber Composite (CFRP)"	0.0032	3.6552e+07
2	"Aluminum Alloy"	0.0032	3.6519e+07
3	"Fiberglass Composite (GFRP)"	0.0089	3.6574e+07
4	"PLA Plastic"	0.0633	3.6486e+07
5	"ABS Plastic"	0.1108	3.6497e+07
6	"Wood (Birch)"	0.0222	3.6497e+07

```
toc
```

Interpretation of Results - Final Recommended Design Solution

Using the results of your thrust-to-weight analysis and finite element analysis, recommend a final drone arm design that meets all required constraints. Your recommendation should clearly identify the selected design (arm geometry and material choice), make use of quantitative results to justify your decisions, demonstrate that the design meets the constraints, explain the trade-offs considered, and include comparisons to alternative designs.

Based on our data, we recommend the usage of chris_final_drone_arm made with birch wood to be used as a drone arm. The design exceeds the minimum 0.5kg payload capacity with a payload capacity of 0.92kg; exceeds the recommended safety factor range of 1.5-2 by having a safety factor of 2.2; and has the lowest material cost of \$0.06 out of our material options, with the average material cost of \$0.3232 and the most expensive material option being \$0.950.95 for a carbon fiber polymer composite. However, the arm has a maximum displacement of 22mm, considerably higher than aluminum alloy and carbon fiber polymer composite's maximum displacement of 3mm. Despite the high displacement, the birch wood drone arm still exceeds the recommended safety factor range, allowing us to still consider the usage of that material for a drone arm.

In contrast, DroneArm_Dogbone made with carbon fiber reinforced polymer was our second-best option: It has a 0.97kg payload capacity, a safety factor of 1.7, costs \$0.49 in material, and has a maximum displacement of 47mm. Despite the slightly higher payload capacity, the dogbone design comes with too many drawbacks compared to chris_drone_arm_v3. The safety factor, maximum displacement, and material cost was considerably worse than chris_final_drone_arm made of birch wood, despite the use of a premium engineering material, so we ruled out DroneArm_Dogbone.

Thus, chris_final_drone_arm made of birch wood is the most feasible drone arm design that we've analyzed. The design's high cost efficiency makes it very suitable to mass produce for low-cost drone units.

What are some limitations of your work?

Generally, simulations can only predict reality through approximations, but cannot accurately reproduce it. In a similar manner, the FEA results will deviate from the drone arm's real-world behavior due to the assumptions and scope we worked with.

For example, the FEA doesn't take the material's manufacturing process into consideration. If the drone arm was made through FDM 3D printing, then its structural characteristics will be anisotropic, while producing the drone arm through subtractive manufacturing techniques like CNC or injection molding would give it isotropic structural characteristics. While it's true that there could also be defects from its manufacturing process that can lower the arm's actual yield strength from the theoretical, taking the drone arm's safety factor into account during our analysis compensates for that. Altogether, our FEA most accurately reflects the structural characteristics of a drone arm not made from FDM 3D printing because our solution doesn't take anisotropy into account.

Additionally, our FEA solution doesn't account for the vibration and variable loading from real-world operating conditions. Throughout its service life, the drone would likely go through hard acceleration from takeoff and corrective maneuvers, subjecting the drone arm to cyclic stresses. The cyclic stresses will cause fatigue and increase the likelihood of failure over time. Meanwhile, propellers can cause resonance from motor harmonics within the drone body during usage. If a propeller vibrates a drone arm close to its natural frequency, it can compound its likelihood for failure, especially if the drone has gone through a substantial number of loading cycles. Because of that, our work is best for predicting a drone arm's short-term performance from a static load and further work will be needed to predict the longer-term durability of our drone arm.

Additionally, the design cost analysis doesn't take the manufacturing process into consideration, but rather the cost of the raw material for the drone arm volume. While it's true that the manufacturing expenses are indirectly reflected in the relative cost differences of each material selection, each of our proposed designs have unique geometry and pose different types of manufacturing challenges that indirectly increase the overall cost of the design. In other words, two different drone arm designs made with the same material with the same volume as each other will likely have different costs because, for example, one design might be more complex to machine than the other.

What are practical next steps?

After the FEA step, we will go back to the CAD phase and model real-world threads and ensure enough clearance for our electronic components. We will also verify that our design would be manufacturable with the manufacturing technique that we've loosely considered while creating our first design drafts. For example, since our drone arm will be made of birch wood, we'll have to verify if we're able to recreate our geometry visualized in the CAD phase given the tooling available for making a wood component.

Next, we'll build real-world prototypes of our drone. Our first prototype will be quickly 3D printed with PLA or ABS plastic or laser cut to verify fitment and clearance of all of our components, but it won't actually be functional. Afterwards, we will build the drone arm out of our chosen material to conduct laboratory tests to verify the FEA results. We will also test for motor harmonics, which our FEA solution doesn't cover.

If real-world tests verify that our drone arm design meets our design specifications with the entire drone, we will create the bill of materials and engineering drawings with GD&T so the drone can be mass-produced by suppliers.